

DATA 3464: Fundamentals of Data Processing

Missing and weird data

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Topic overview

- What to do with missing data
- Detecting and handling outliers

Resources used:

- [Feature Engineering Chapter 8](#)
- Hands on Machine Learning with Scikit-Learn and Tensorflow/PyTorch, Chapter 2.
Available at [MRU Library](#)
- [Scikit-learn user guide: Chapters 2 and 7](#)

The problem

- As you've seen, real-world data is messy
- Missing values are common, other values don't make sense
- We need to decide how to deal with these problems

What examples have we seen so far?

Why might data be missing or weird?*

*I'm using "weird" as an informal catch-all for unexpected or outlier values

Missing data

When data are missing in the *features* we have a few options:

0. Do nothing! Some algorithms (e.g. [Decisions Trees](#)) can handle missing values
1. Remove *features* with missing values
2. Remove *samples* with missing values
3. Invent a new value to represent "missingness"
4. **Impute** a value based on other data

Most important: understand why data are missing (more EDA!)

Option 1: removing features

- If a feature has:
 - A high proportion of missing values, *and*
 - Little apparent relationship to the target *or*
 - its information is redundant with other features
- It may be reasonable to remove it entirely. You can drop it, e.g.:

```
df.drop(columns=['feature_name'], inplace=True)
```

or (probably more reliable) just not select it when building your pipeline

Option 2: removing samples

- If only a small number of samples have missing values, you can drop them **from the training data**:

```
train.dropna(subset=[["features", "we", "care", "about"]], inplace=True)
```

- Good idea if the same samples have missing values from multiple features
- Still useful to explore *why* data are missing

What should we do for inference?

Option 3: invent a new value

- Categorical features: add a new category for "missing"
- Add a new binary feature indicating whether the value was missing
- I have seen advice to use extreme values for numerical features, like the -1 income in the OKCupid dataset, but I'm not convinced this is a good idea

Case study: where missingness is informative

Option 4: impute missing values

- Fill in the missing values with an "educated guess"
- Replace missing value with:
 - constant
 - mean, median, or mode (`most_frequent`)
- Use other features to infer missing value:
 - K-nearest neighbours
 - simple models to predict missing values
- Can be combined with option 3 to indicate missing features
- How much to impute? [Feat.Engineering](#) suggests no more than 20%

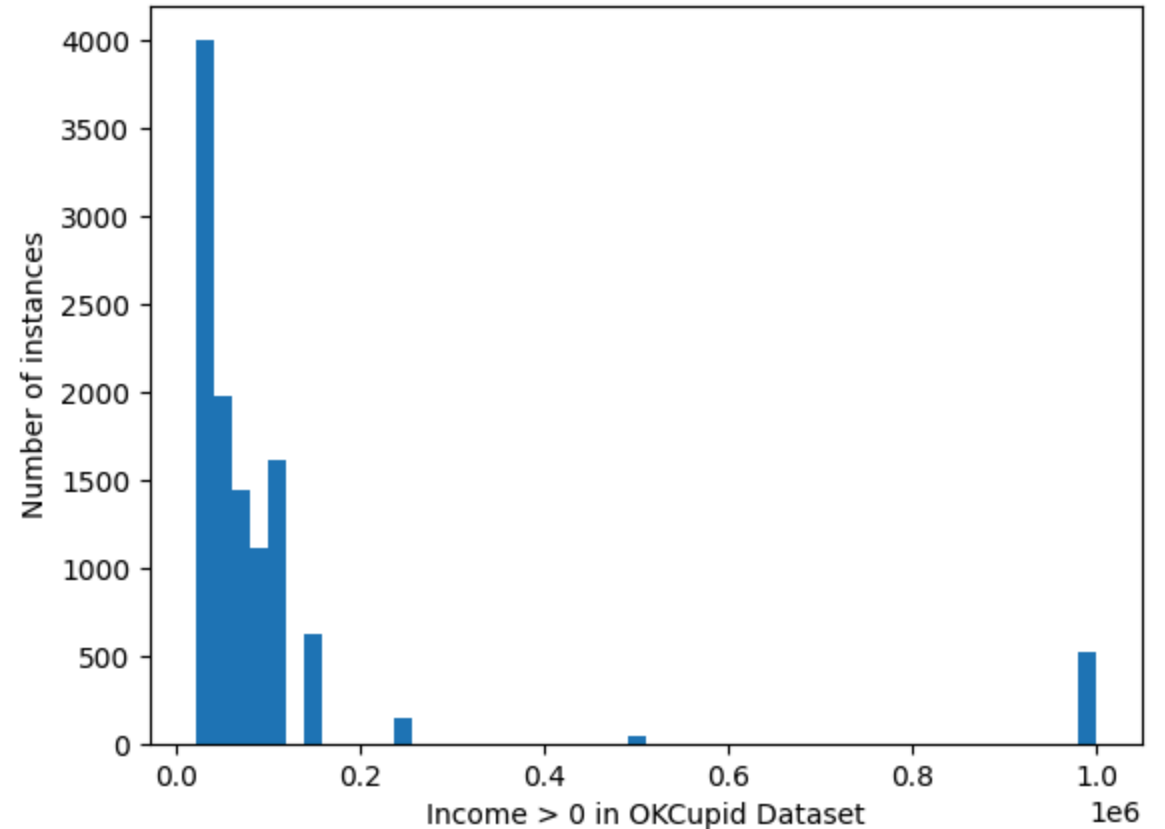
Outliers

An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism. -- D. M. Hawkins

- There are many entire books dedicated to outlier detection
- Useful for anomaly detection, e.g:
 - fraud detection
 - failure prediction
 - **Many other applications**
- Our focus is on dealing with outliers in preprocessing

Detecting outliers

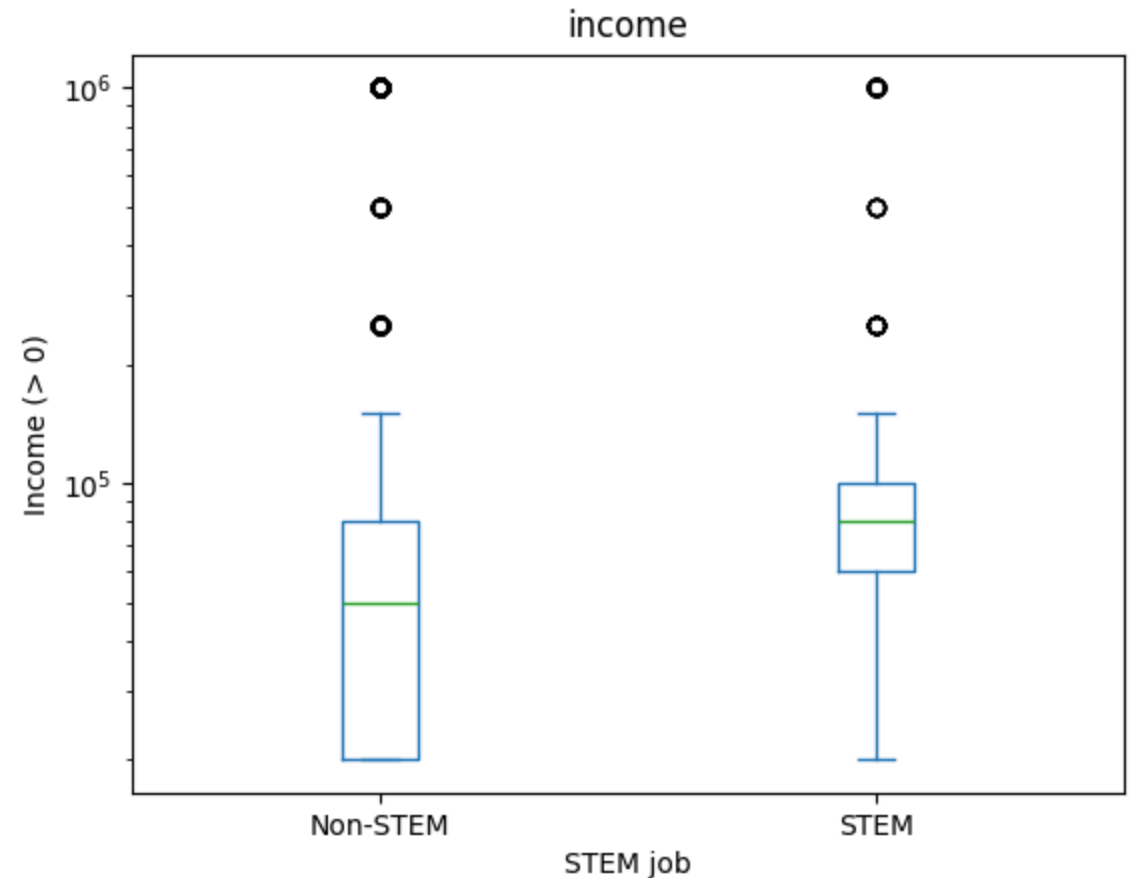
- Visually as part of EDA
- Statistically, e.g. $> 3\sigma$
- Algorithmically, e.g. [Isolation Forests](#)
- As usual, context and domain knowledge are essential



Context matters!

- Look at the relationship between outliers and target (training dataset, of course)

What do the dots on box plots represent?



What to do about outliers?

- Data transformations
- Drop the samples
- Encode them somehow
- Leave them alone

As usual, very data- and model-dependent. Tree based methods are particularly impacted by outliers!

Any other ideas?